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1. An ATM allows a user to perform multiple withdrawal transactions. Write a C program that repeatedly asks the user to enter a withdrawal amount. Continue taking amounts while the entered amount is greater than 0. For every valid withdrawal, subtract the amount from an initial balance of 50,000. At the end, display the remaining balance and the number of withdrawals.

```

C atm.c X C stdntmarks.c C recharge.c C shopping.c C bill2.c
folderofc.c > C atm.c
1  #include<stdio.h>
2  int main(){
3      int amount = 50000;
4      int withdrawal,counter=0;
5
6      do{
7          printf("Enter the withdrawal amount");
8          scanf("%d",&withdrawal);
9
10         amount = amount-withdrawal;
11         counter++;
12     }
13     while(withdrawal>0);
14     printf("Remaining amount is %d\n",amount);
15     printf("Total number of withdrawals are %d",counter);
16 }
17

```

```

Enter the withdrawal amount1200
Enter the withdrawal amount2500
Enter the withdrawal amount0
Remaining amount is 46300
Total number of withdrawals are 3
PS C:\Users\I C\VS\folderofc.c>

```

2. A teacher wants to calculate the total and average marks of students. Write a C program that repeatedly asks the user to enter marks between 0 and 100. Continue taking marks while the entered value is valid. When the user enters -1, stop the process and display the total marks, number of students, and average marks

```
C atm.c C stdntmarks.c X C recharge.c C shopping.c C bill
folderofc.c > C stdntmarks.c
1  #include<stdio.h>
2  int main(){
3      int marks,students=0,total_marks=0;
4      float avg;
5      do{
6          printf("Enter marks(0-100): ");
7          scanf("%d",&marks);
8
9          total_marks = total_marks+marks;
10         students++;
11     }
12     while(marks>-1);
13     avg = total_marks/students;
14     printf("The avg is: %.2f\n",avg);
15     printf("The total marks are: %d\n",total_marks);
16     printf("The number of students are: %d\n",students);
17 }
```

```
Enter marks(0-100): 60
Enter marks(0-100): 26
Enter marks(0-100): 70
Enter marks(0-100): 70
Enter marks(0-100): -1
The avg is: 45.00
The total marks are: 225
The number of students are: 5
PS C:\Users\I C\VS\folderofc.c>
```

3. A mobile user wants to add balance several times. Write a C program that repeatedly asks for a recharge amount. Continue while the amount is greater than 0. Add every recharge amount to the total balance. If the total recharge becomes greater than 5,000, display "Recharge Limit Reached" and stop. At the end, display the total recharged amount and number of recharge attempts.

```
folderofc.c > C recharge.c
1  #include <stdio.h>
2
3  int main() {
4      int re_amount, total = 0, attempts = 0;
5
6      do {
7          printf("Enter Amount Of Recharge: ");
8          scanf("%d", &re_amount);
9
10         total = total + re_amount;
11         attempts++;
12         if (total > 5000){
13             printf("Recharge Limit Reached\n");
14             break;
15         }
16     }
17     while (re_amount > 0);
18     printf("The total recharged amount is: %d\n", total);
19     printf("The number of recharge attempts are: %d\n", attempts);
20
21     return 0;
22 }
```

```
Enter Amount Of Recharge: 2500
Enter Amount Of Recharge: 1200
Enter Amount Of Recharge: 0
The total recharged amount is: 3700
The number of recharge attempts are: 3
PS C:\Users\I C\VS\folderofc.c> |
```

4. You are shopping online and want to purchase multiple items. Write a C program that asks the user to enter the price of an item and then asks whether they want to add another item (1 = Yes, 0 = No). Add the price of every item to the total. After shopping is completed, if the total is greater than 10,000, apply a 10% discount. Display the total price, discount, and final amount.

```
atm.c  stdntmarks.c  recharge.c  shopping.c X  bill2.c  pin.c
folderofc.c > shopping.c
1  #include<stdio.h>
2  int main(){
3      int price,again;
4      float discount=0,final_amount;
5
6      do{
7          printf("Enter the price: ");
8          scanf("%d",&price);
9
10         final_amount = final_amount+price;
11
12         printf("Do you want to add another item(1 = Yes, 0 = No): ");
13         scanf("%d",&again);
14     }
15     while(again==1);
16
17     if(final_amount>10000)
18         discount = final_amount*0.10;
19     else
20         discount = 0;
21     float total_price = final_amount-discount;
22     printf("The total price is: %.2f\n",final_amount);
23     printf("The discount is: %.2f\n",discount);
24     printf("The final amount is %.2f\n",total_price);
25 }
26
```

```
Enter the price: 5000
Do you want to add another item(1 = Yes, 0 = No): 1
Enter the price: 5500
Do you want to add another item(1 = Yes, 0 = No): 0
The total price is: 10500.00
The discount is: 1050.00
The final amount is 9450.00
```

5. A customer orders food from a restaurant. Write a C program that asks for the price of each item and whether the customer wants to order another item (1 = Yes, 0 = No). Add all item prices to calculate the total bill. If the bill is greater than 5,000, apply a 5% discount. Display the total bill, discount, and final bill.

```
atm.c  stdntmarks.c  recharge.c  shopping.c  bill2.c X  pin.c
folderofc.c > C bill2.c
1  #include<stdio.h>
2  int main(){
3      int price,again;
4      float discount=0,final_amount;
5
6      do{
7          printf("Enter the price: ");
8          scanf("%d",&price);
9
10         final_amount = final_amount+price;
11
12         printf("Do you want to add another item(1 = Yes, 0 = No): ");
13         scanf("%d",&again);
14     }
15     while(again==1);
16
17     if(final_amount>5000)
18         discount = final_amount*0.05;
19     else
20         discount = 0;
21     float total_price = final_amount-discount;
22     printf("The total price is: %.2f\n",final_amount);
23     printf("The discount is: %.2f\n",discount);
24     printf("The final amount is %.2f\n",total_price);
25 }
```

```
Enter the price: 1500
Do you want to add another item(1 = Yes, 0 = No): 1
Enter the price: 2500
Do you want to add another item(1 = Yes, 0 = No): 1
Enter the price: 3000
Do you want to add another item(1 = Yes, 0 = No): 0
The total price is: 7000.00
The discount is: 350.00
The final amount is 6650.00
```

6. A user has a maximum of 3 login attempts. Write a C program that asks the user to enter a PIN. The correct PIN is 1234. For every incorrect PIN, increase the attempt count by 1 and display the remaining attempts. If the correct PIN is entered, display "Login Successful". If all three attempts are incorrect, display "Account Locked".

```
atm.c  stdntmarks.c X  recharge.c  shopping.c  bill2.c
folderofc.c > C pin.c
1  #include <stdio.h>
2  int main() {
3      int pin = 1234, enter_pin;
4      for (int i=1; i<=3;i++){
5
6          printf("Enter the pin: ");
7          scanf("%d", &enter_pin);
8
9          if(enter_pin == pin){
10             printf("Login Successful\n");
11             break;
12         }
13         else{
14             printf("Incorrect PIN\n");
15             printf("Remaining attempts: %d\n", 3 - i);
16         }
17         if(i == 3){
18             printf("Account Locked\n");
19         }
20     }
21 }
```

```
Enter the pin: 1345
Incorrect PIN
Enter the pin: 5678
Incorrect PIN
Enter the pin: 678
Incorrect PIN
Account Locked
PS C:\Users\I C\VS\folderofc.c>
```

7. A teacher wants to store the marks of 5 students in an array. Write a C program to take the marks and store them in the array. After taking all marks, use a loop to calculate the total marks, average marks, highest marks, and lowest marks.

```
stdntmarks.c  recharge.c  shopping.c  bill2.c  pin.c  marks.c
folderofc.c > marks.c
1  #include <stdio.h>
2  int main() {
3      int marks[5],total = 0;
4      float avg,highest,lowest;
5
6      for (int i=0;i<5;i++){
7          printf("Enter marks: ");
8          scanf("%d",&marks[i]);
9
10         total=total + marks[i];
11
12         if(i==0){
13             highest = marks[i];
14             lowest = marks[i];
15         }
16         else{
17             if (marks[i] > highest){
18                 highest = marks[i];
19             }
20
21             if (marks[i] < lowest){
22                 lowest = marks[i];
23             }
24         }
25     }
26     avg = total/5;
27     printf("The total marks are: %d\n", total);
28     printf("The average is: %.2f\n", avg);
29     printf("The highest marks are: %.0f\n", highest);
30     printf("The lowest marks are: %.0f\n", lowest);
31 }
```

```
Enter marks: 50
Enter marks: 70
Enter marks: 0
Enter marks: 80
Enter marks: 12
The total marks are: 212
The average is: 42.00
The highest marks are: 80
The lowest marks are: 0
```

8. An online shopping system stores the prices of 5 products in an array. Write a C program to take the prices from the user and use a loop to calculate the total price. If the total is greater than 10,000, apply a 10% discount. Display the original total, discount, and final amount.

```
stdntmarks.c  recharge.c  shopping.c  bill2.c  pin.c  ma
folderofc.c > C shopping2.c
1  #include<stdio.h>
2  ∨ int main(){
3      int price[5],original_total=0;
4      float discount,final_amount;
5
6  ∨  for(int i=0;i<5;i++){
7      printf("Enter price: ");
8      scanf("%d",&price[i]);
9
10     original_total = original_total+price[i];
11 }
12 ∨ if(original_total>10000)
13     discount = original_total*0.10;
14 ∨ else
15     discount = 0;
16 final_amount = original_total-discount;
17 printf("The original total is: %d\n",original_total);
18 printf("The Discount is: %.2f\n",discount);
19 printf("The final amount is: %.2f\n",final_amount);
20 }
21
```

```
Enter price: 5000
Enter price: 1200
Enter price: 2500
Enter price: 5690
Enter price: 234
The original total is: 14624
The Discount is: 1462.40
The final amount is: 13161.60
PS C:\Users\I C\VS\folderofc.c>
```

9. An electricity company wants to store the electricity units consumed by 5 households in an array. Write a C program to take the units for all 5 households and use a loop to calculate the total units consumed, highest units, and lowest units. For each household, calculate the bill using 10 rupees per unit. If a household consumes more than 500 units, add an additional 5% surcharge to its bill. Finally, display the total units consumed and total amount collected.

[C atm.c](#)[C stdntmarks.c](#)[C recharge.c](#)[C shopping.c](#)[C bill2.c](#)[pin.c](#)folderofc.c > [C electricity.c](#)

```
1  #include <stdio.h>
2  int main() {
3      int units[5], totalUnits = 0, highest, lowest;
4      float bill, totalAmount = 0;
5
6      for (int i = 0; i < 5; i++) {
7          printf("Enter units for household %d: ", i + 1);
8          scanf("%d", &units[i]);
9
10         totalUnits = totalUnits + units[i];
11         if (i == 0){
12             highest = units[i];
13             lowest = units[i];
14         }
15         else{
16             if (units[i] > highest) {
17                 highest = units[i];
18             }
19
20             if (units[i] < lowest) {
21                 lowest = units[i];
22             }
23         }
24         bill = units[i] * 10;
25         if (units[i] > 500){
26             bill = bill + (bill * 0.05);
27         }
28         totalAmount = totalAmount + bill;
29     }
30     printf("\nTotal units consumed: %d\n", totalUnits);
31     printf("Highest units: %d\n", highest);
32     printf("Lowest units: %d\n", lowest);
33     printf("Total amount collected: %.2f\n", totalAmount);
34 }
```

```
Enter units for household 1: 120
Enter units for household 2: 112
Enter units for household 3: 90
Enter units for household 4: 20
Enter units for household 5: 91
```

```
Total units consumed: 433
Highest units: 120
Lowest units: 20
Total amount collected: 4330.00
PS C:\Users\I C\VS\folderofc.c> █
```